

softstart_routed

Design Report

Rev 1 | 2026-06-16 | jlcpcb-tier1

kicad-tools 0.13.0

Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	78 (56 SMD, 18 THT, 4 other)
Nets	41
Traces	4727 segments
Vias	158
Board Size	150.0 x 100.0 mm

Design Overview

Theory of Operation

Generator Soft-Start - Supercapacitor Power Assist

120VAC soft-start for 8000 BTU AC on Honda EU1000i

STM32G031K8T6 MCU, 2x30S supercap banks, back-to-back FETs + UCC27211 drivers

Power Architecture

Power Rails: +3V3, GND, PWR_FLAG

Regulator	Device
U4	XC6206-3.3V
U9	LM7812

Assembly Notes

1 fine-pitch component; 6 polarized components

- **Fine-pitch components:** 1 (U1)
- **Polarized components:** 6 -- check orientation markings

Hand-Solder (THT) Components

The following 18 through-hole components are **excluded from the SMT pick-and-place file** and must be hand-soldered (or wave/selective-soldered) after SMT assembly. They appear in the BOM for sourcing.

Value	Package	Qty	References
DB107	Diode_Bridge_DIP-4_W 7.62mm_P5.08mm	1	D1
15A	Fuseholder_Cylinder-5x 20mm_Schurter_0031.8 201_Horizontal_Open	1	F1
TB	TerminalBlock_bornier-2 _P5.08mm	4	J1, J2, J3, J4
SWD	PinHeader_1x06_P2.54 mm_Vertical	1	J5
IRFB4110	TO-220-3_Vertical	4	Q1A, Q1B, Q2A, Q2B
150R 5W	R_Axial_DIN0617_L17.0 mm_D6.0mm_P25.40m m_Horizontal	1	R6
100R 5W	R_Axial_DIN0617_L17.0 mm_D6.0mm_P25.40m m_Horizontal	2	R20, R21
275VAC	RV_Disc_D12mm_W4.2 mm_P7.5mm	1	RV1
RESET	SW_PUSH_6mm	1	SW1
H11AA1	DIP-6_W7.62mm	1	U2
LM7812	TO-220-3_Vertical	1	U9

Off-board Assemblies

The following subsystems are part of the design but are **not placed on the PCB** -- they connect through board connectors and are assembled by hand (DNP for fab assembly).

SUPERCAP_BANK_POS

Positive half-cycle supercapacitor bank (off-board, 30S string + per-cell balance resistors)

Property	Value
Board connector	J3
Part	Abracon ADCR-S03R0SA256MB + 30x 3.3k 1/4W balance resistors
Quantity	30
Voltage	75V charge target (90V stack rating)
Capacitance	0.83F
Assembly	hand solder (DNP for fab assembly)

Wiring: J3 pin 1 -> SCAP_POS+ (bank positive terminal); J3 pin 2 -> SCAP_POS_GND (bank return, star ground). 30 cells in series, 25F 3.0V each, one 3.3k resistor soldered lead-to-lead across each cell.

SUPERCAP_BANK_NEG

Negative half-cycle supercapacitor bank (off-board, 30S string + per-cell balance resistors)

Property	Value
Board connector	J4
Part	Abracon ADCR-S03R0SA256MB + 30x 3.3k 1/4W balance resistors
Quantity	30
Voltage	75V charge target (90V stack rating)
Capacitance	0.83F
Assembly	hand solder (DNP for fab assembly)

Wiring: J4 pin 1 -> SCAP_NEG+ (bank positive terminal); J4 pin 2 -> SCAP_NEG_GND (bank return, star ground). 30 cells in series, 25F 3.0V each, one 3.3k resistor soldered lead-to-lead across each cell.

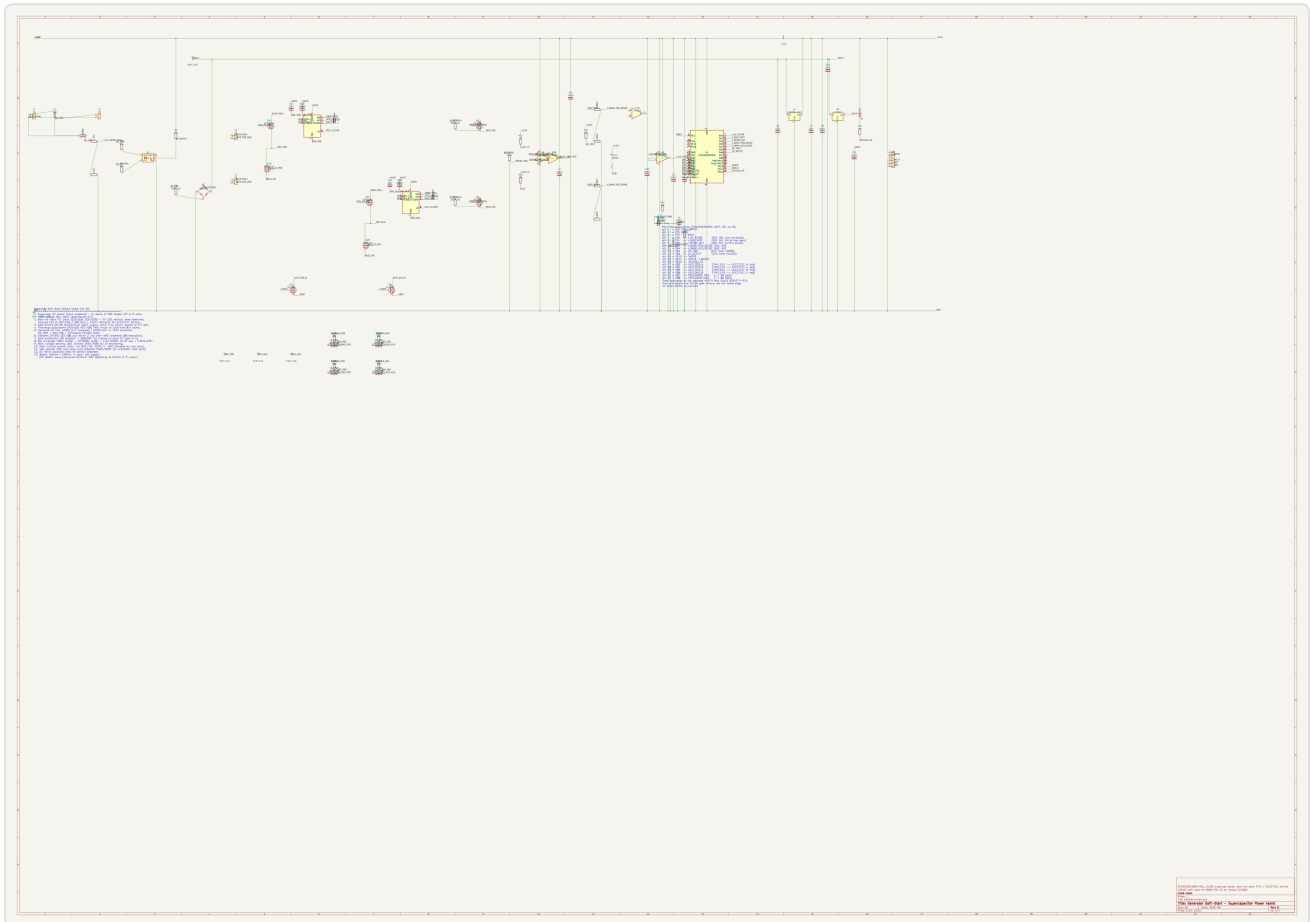
ERC Status

Metric	Count
Errors	0
Warnings	0

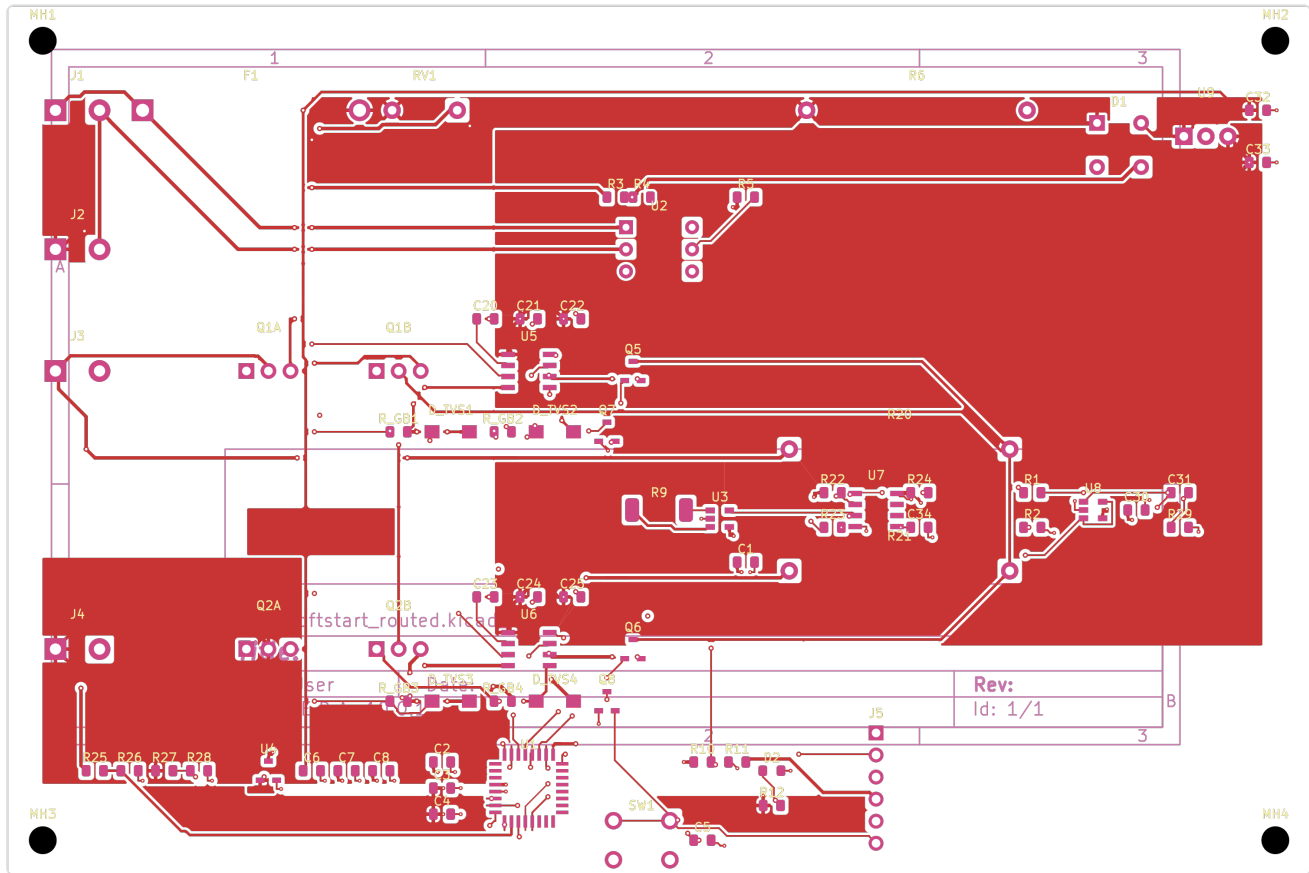
Status: SKIPPED -- ERC skipped by user request

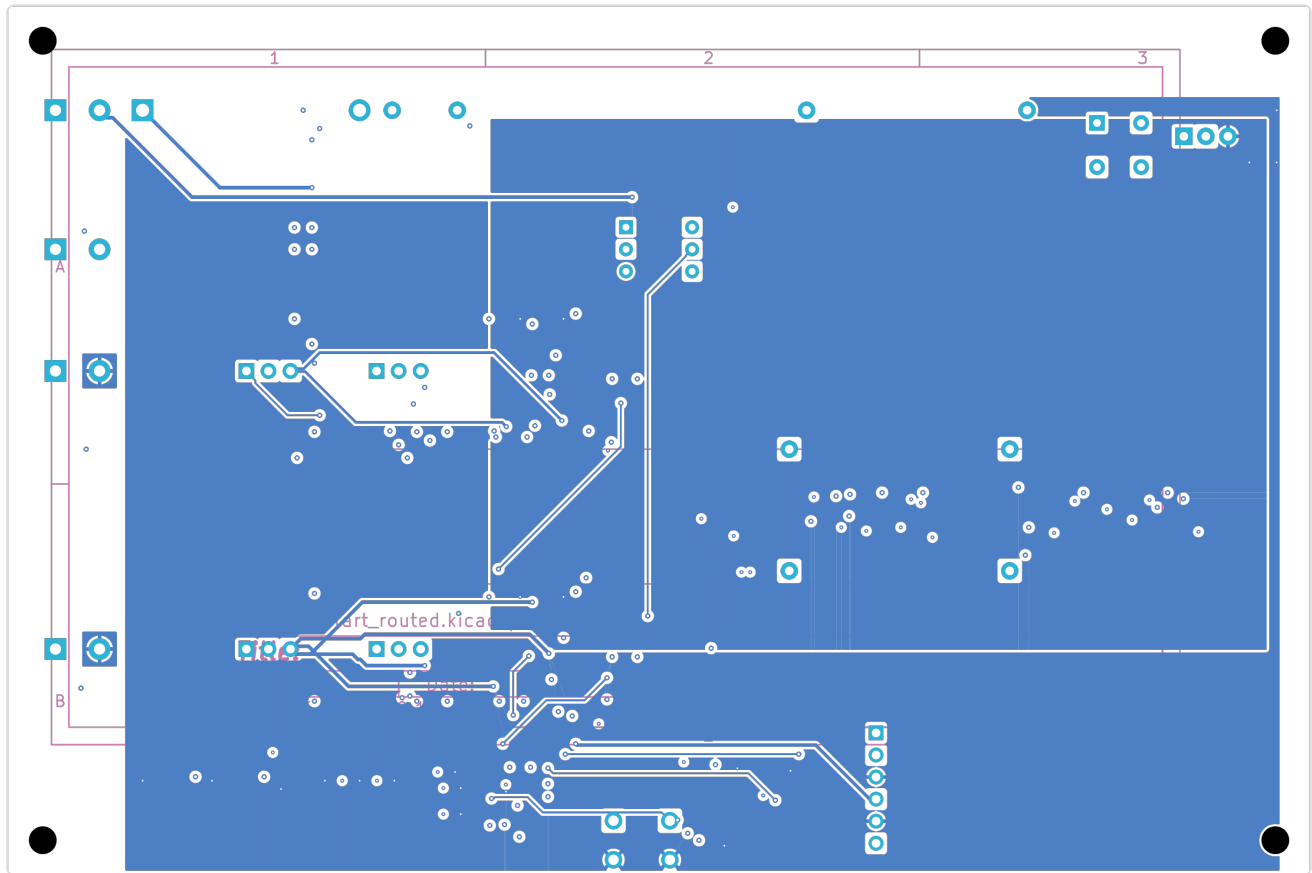
Schematic Overview

Schematic: softstart

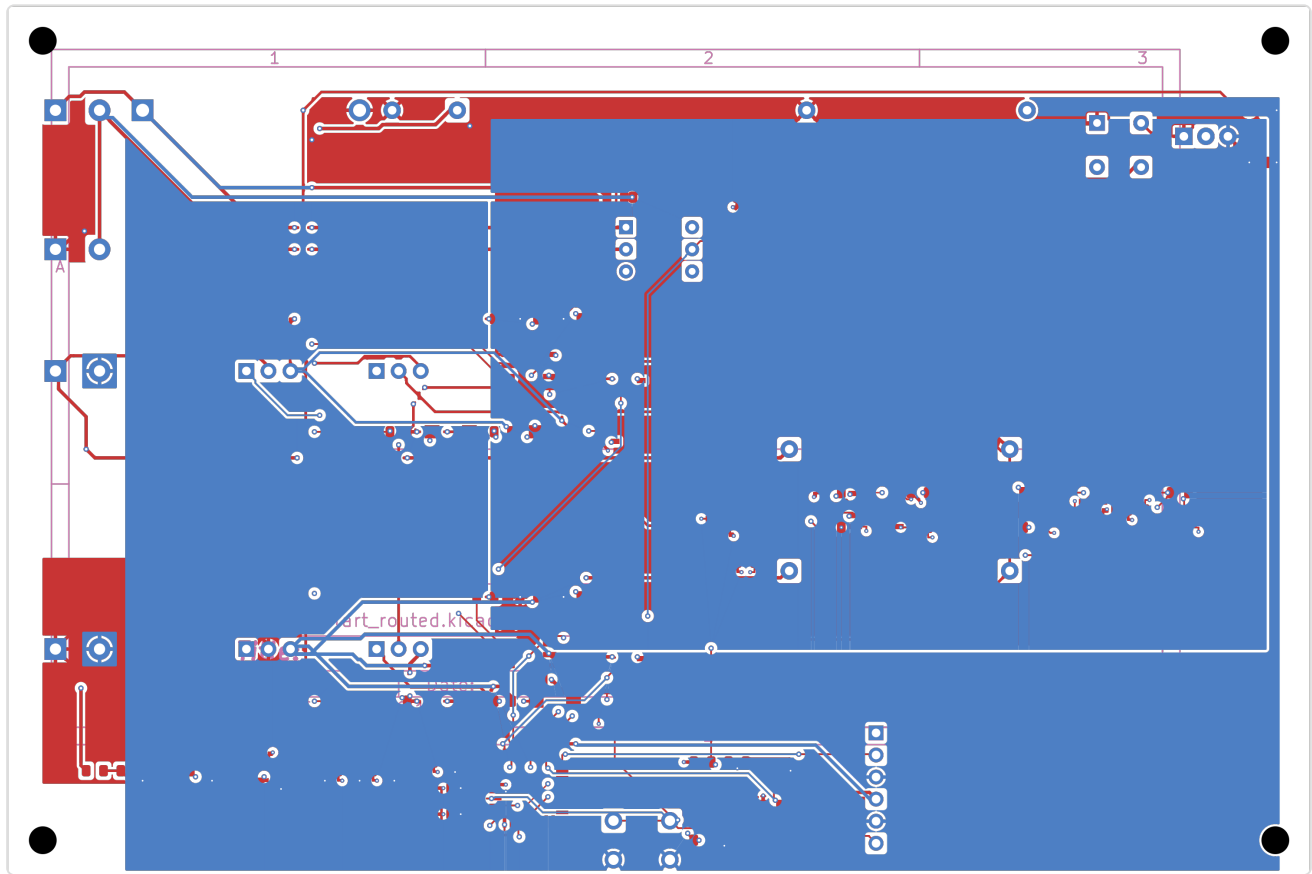


PCB Layout

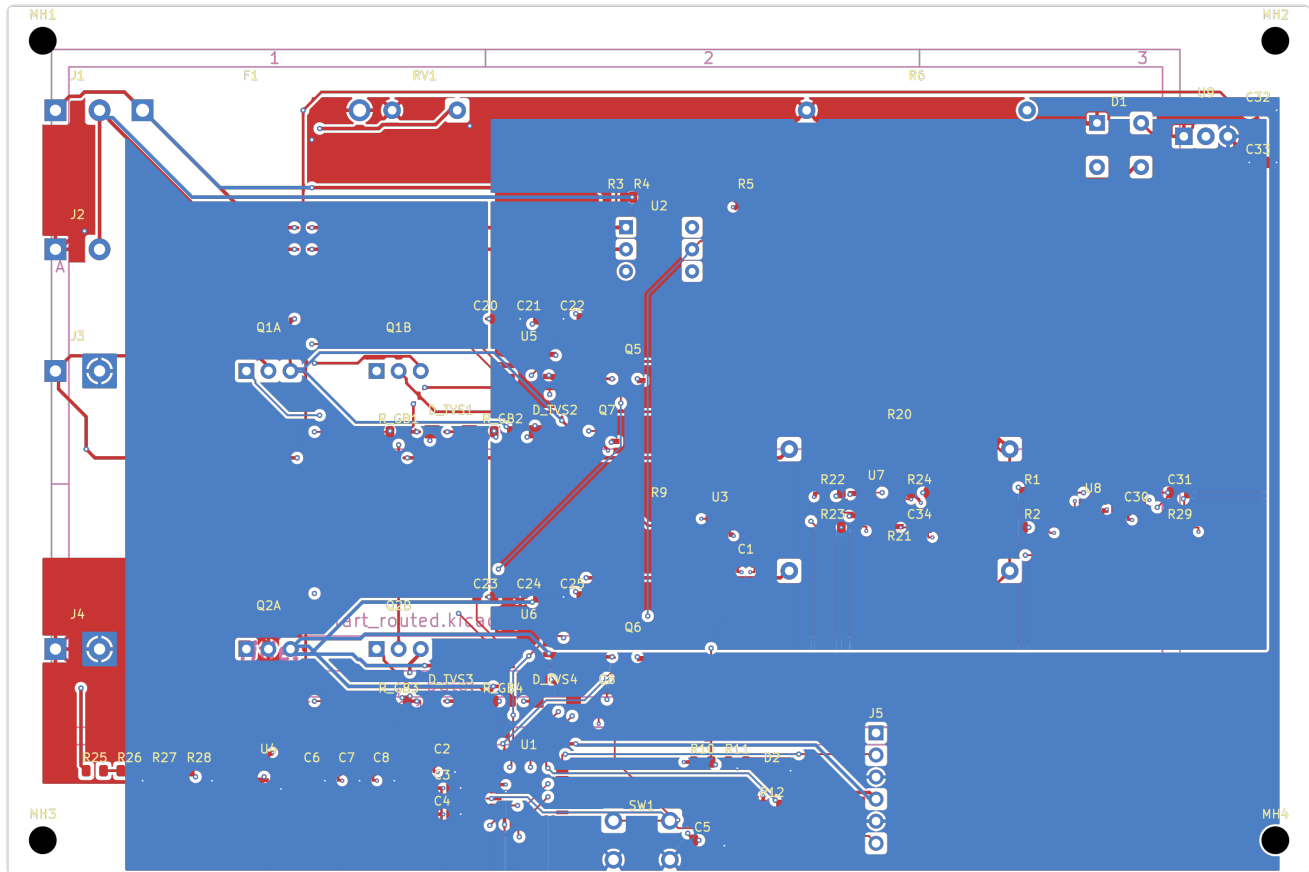




Copper

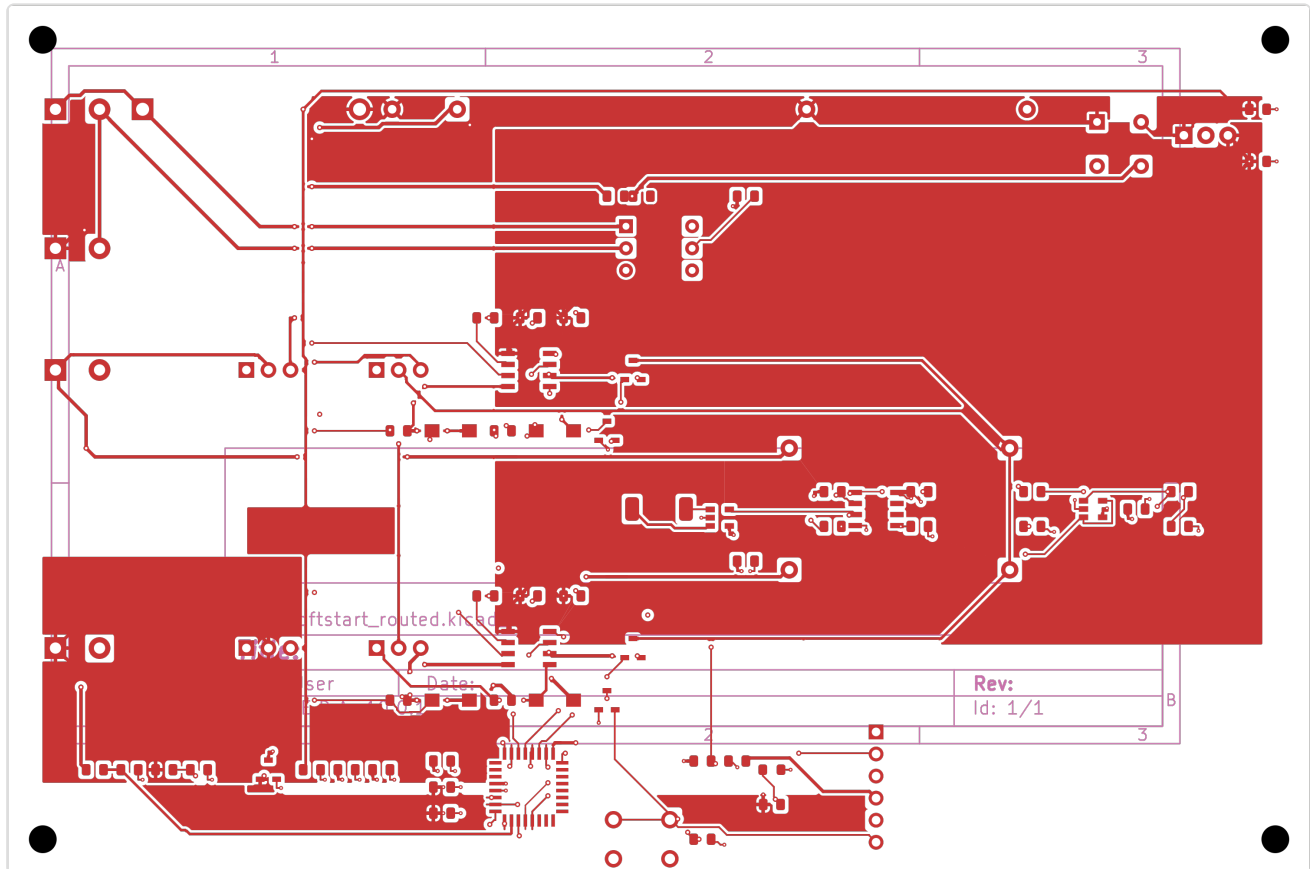


Assembly

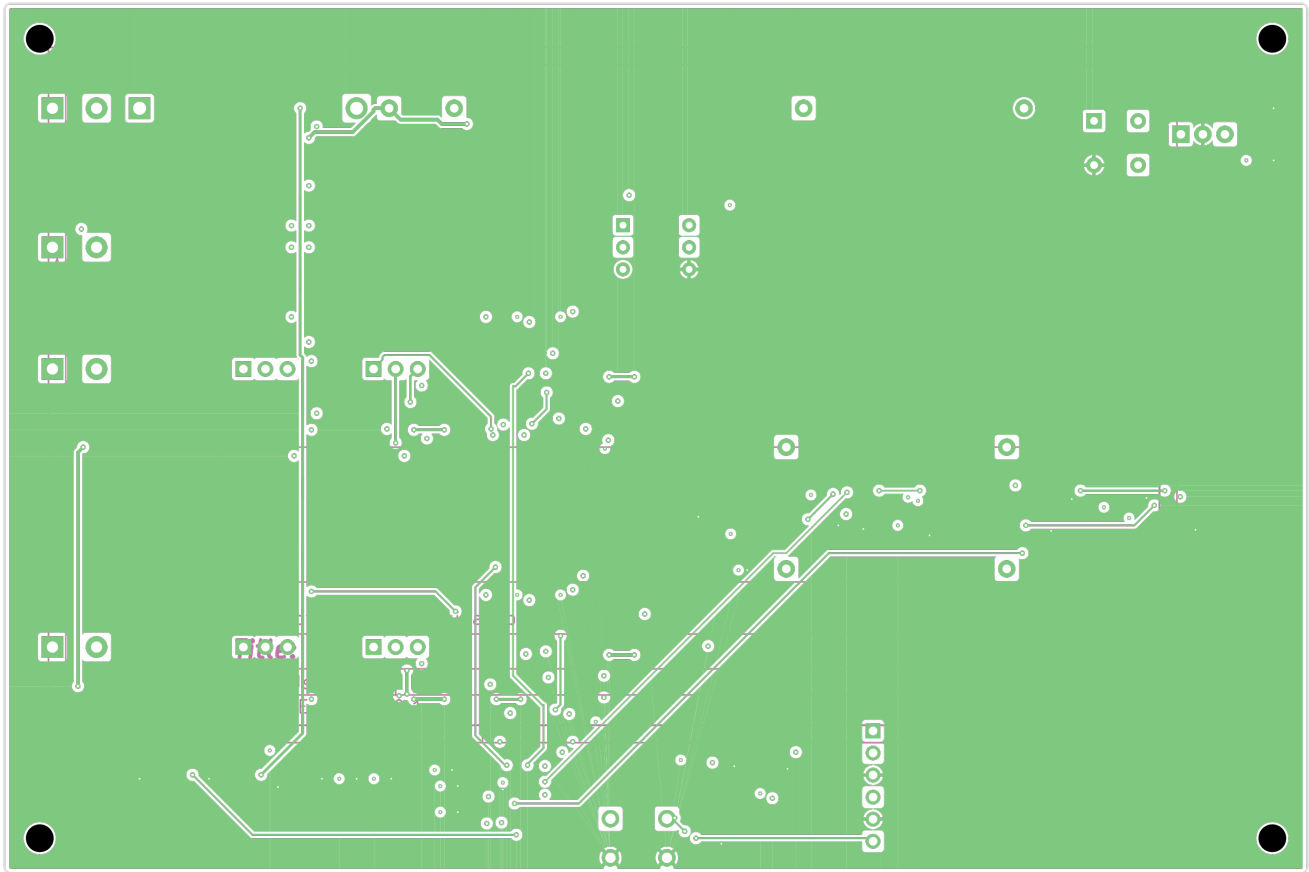


Copper Layers

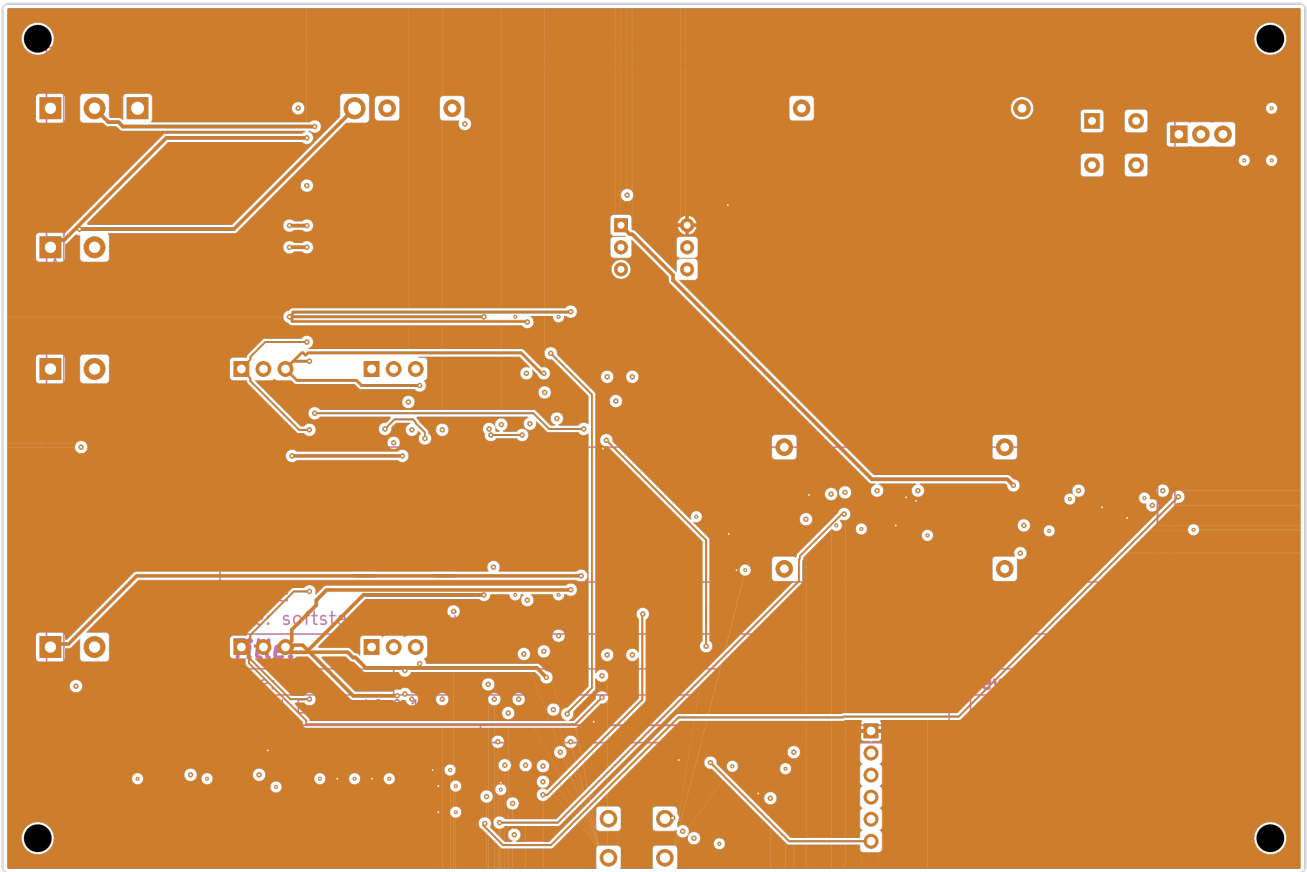
F.Cu



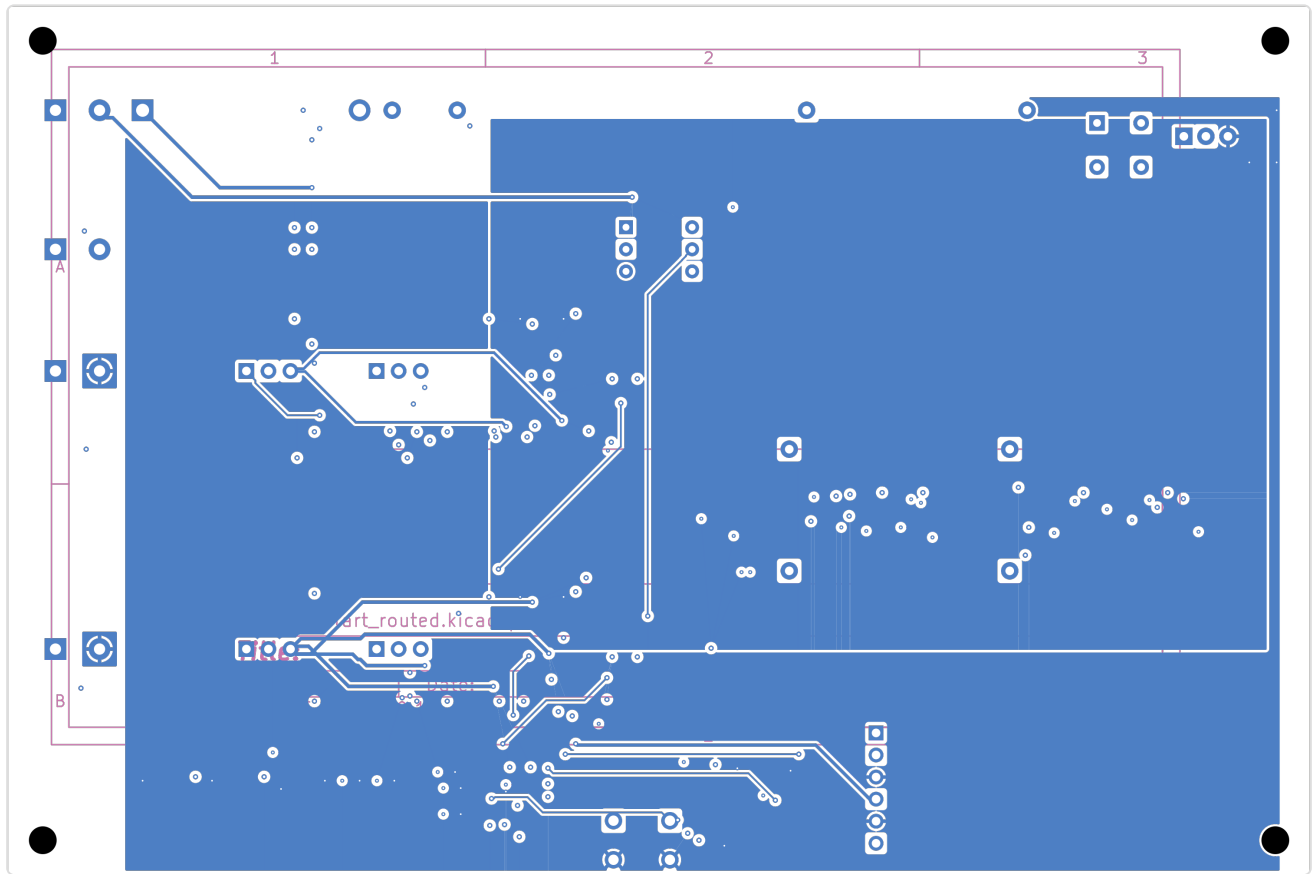
In1.Cu



In2.Cu



B.Cu



Bill of Materials

Value	Package	Qty	References	MPN
100nF		5	C5, C20, C22, C23, C25	
100nF	C_0805_2012Metric	8	C1, C2, C3, C8, C30, C31, C33, C34	
10uF		2	C21, C24	
10uF	C_0805_2012Metric	3	C6, C7, C32	
4.7uF	C_0805_2012Metric	1	C4	
DB107	Diode_Bridge_DIP-4_W7.62mm_P5.08mm	1	D1	
LED		1	D2	
SMBJ18A	D_SMA	4	D_TVS1, D_TVS2, D_TVS3, D_TVS4	
15A		1	F1	
SWD		1	J5	
TB	TerminalBlock_bornier-2_P5.08mm	4	J1, J2, J3, J4	
2N7002	SOT-23	2	Q7, Q8	
AO3400	SOT-23	2	Q5, Q6	
IRFB4110	TO-220-3_Vertical	4	Q1A, Q1B, Q2A, Q2B	
100R 5W	R_Axial_DIN0617_L17.0mm_D6.0mm_P25.40mm_Horizontal	2	R20, R21	
10k		5	R2, R10, R11, R23, R24	
10k	R_0805_2012Metric	6	R_GB1, R_GB2, R_GB3, R_GB4, R5, R29	
150R 5W		1	R6	

Value	Package	Qty	References	MPN
	R_Axial_DIN0617_ L17.0mm_D6.0mm _P25.40mm_Horiz ontal			
1M		1	R1	
1k		2	R12, R22	
270k		2	R25, R27	
275VAC	RV_Disc_D12mm_ W4.2mm_P7.5mm	1	RV1	
33k	R_0805_2012Metri c	2	R3, R4	
5mR	R_2512_6332Metri c	1	R9	
9.1k		2	R26, R28	
RESET		1	SW1	
H11AA1	DIP-6_W7.62mm	1	U2	
INA180A3	SOT-23-5	1	U3	
LM393	SOIC-8_3.9x4.9m m_P1.27mm	3	U7, U7, U7	
LM7812	TO-220-3_Verical	1	U9	
MCP6001	SOT-23-5	1	U8	
STM32G031K8T6	LQFP-32_7x7mm_ P0.8mm	1	U1	
UCC27211	SOIC-8_3.9x4.9m m_P1.27mm	2	U5, U6	
XC6206-3.3V		1	U4	
Abracon ADCR- S03R0SA256MB + 30x 3.3k 1/4W balance resistors	Off-board via J3 (hand solder, DNP for fab assembly)	30	SUPERCAP_BANK _POS	Abracon ADCR- S03R0SA256MB + 30x 3.3k 1/4W balance resistors
Abracon ADCR- S03R0SA256MB +	Off-board via J4 (hand solder, DNP for fab assembly)	30	SUPERCAP_BANK _NEG	Abracon ADCR- S03R0SA256MB +

Value	Package	Qty	References	MPN
30x 3.3k 1/4W balance resistors				30x 3.3k 1/4W balance resistors

DRC Status

Metric	Count
Errors	10
Warnings	0
Blocking	10

Status: FAIL

Violations by Type

Violation Type	Count
kicad-cli:starved_thermal	3
kicad-cli:clearance	2
kicad-cli:shorting_items	2
kicad-cli:solder_mask_bridge	2
kicad-cli:hole_clearance	1

Manufacturing Readiness

Verdict: NOT_READY

Action Items

- **[CRITICAL]** Fix 10 blocking DRC violations (kicad-cli: starved_thermal (3), clearance (2), shorting_items (2))
- **[OPTIONAL]** Verify zone fill in KiCad for 12 zone-connected nets
- **[OPTIONAL]** Analog-sensitive: U3 (INA180A3) — instrumentation amplifier (TI INA); manual layout review recommended
- **[OPTIONAL]** Analog net: AC_LINE — audio signal; keep short, away from digital/switching nets
- **[OPTIONAL]** Analog net: BUS_LINE — audio signal; keep short, away from digital/switching nets
- **[OPTIONAL]** Analog net: FUSED_LINE — audio signal; keep short, away from digital/switching nets
- **[OPTIONAL]** Analog net: ISENSE_NEG — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: ISENSE_POS — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: I_SENSE_OUT — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: V_AC_SENSE — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: V_AC_SENSE_RAW — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: V_BANK_NEG_SENSE — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: V_BANK_POS_SENSE — analog signal; noise-sensitive, avoid crossing digital signals

Analog Components

1 analog-sensitive component detected -- manual layout review recommended.

Reference	Value	Reason
U3	INA180A3	instrumentation amplifier (TI INA)

Routing Status

Metric	Value
Signal Net Completion	100.0% (27/27)
Overall Completion	100.0%
Complete Nets	41 / 41
Zone-Connected Nets	14
Single-Pad Nets	2 (no routing needed)
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- +3.3V
- AC_LINE
- AC_NEUTRAL
- BUS_LINE
- FUSED_LINE
- GND
- SCAP_NEG+
- SCAP_NEG_GND
- SCAP_POS+
- SCAP_POS_GND
- SRC_NEG
- SRC_POS
- VGATE
- VRECT

Single-Pad Nets

2 single-pad nets (no routing needed) -- not listed individually.

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~5.0 USD
Components (estimated)	~5.72 USD
Assembly (estimated)	~0.0 USD
Total (estimated)	~10.72 USD
Batch Quantity	5
Batch Total (estimated)	~53.6 USD